SOLUSI LATIHAN LOGIC GATES AND COMBINATIONAL CIRCUITS

DDAK 2022/2023

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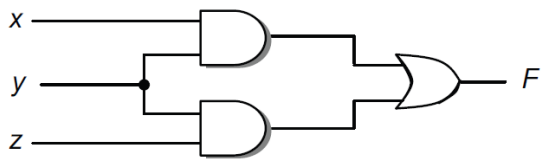
PERHATIAN
- Jangan baca solusi sebelum mengerjakan soal pada dokumen latihan
- Jika ada hal yang kurang jelas dari jawaban harap email ke: **herik@cs.ui.ac.id**

PART I

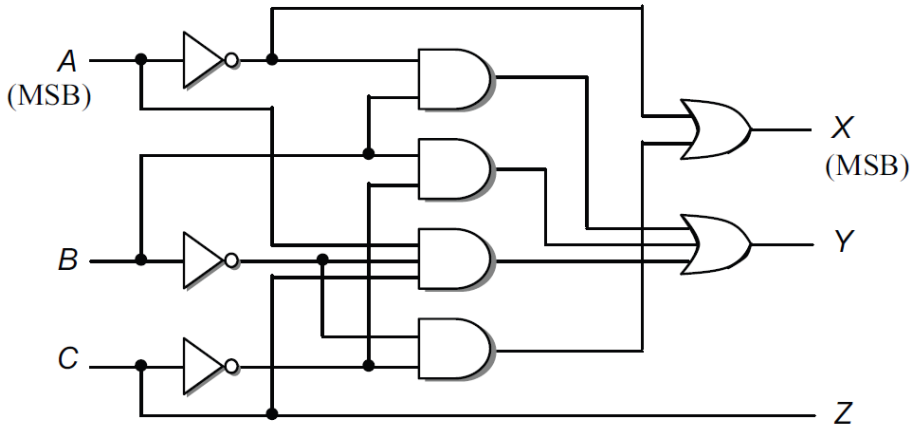
4-1. Draw a logic circuit for the expression $B{\cdot}D + B{\cdot}E + D'{\cdot}F$.

4-2. Draw a logic circuit for the expression $A'{\cdot}B{\cdot}C + B'{\cdot}C{\cdot}D + B{\cdot}C'{\cdot}D + A{\cdot}B{\cdot}D'$.

4-3. What is the Boolean expression for F in the logic circuit below? Express it in SOP form.



4-4. What is the function of the circuit below?



a. Convert from 3-bit sign-magnitude code to 3-bit excess-4 code
b. Convert from 3-bit excess-4 code to 3-bit sign-magnitude code
c. Convert from 3-bit 2's complement number to 3-bit excess-4 code
d. Convert from 3-bit excess-4 code to 3-bit 2's complement number
e. Convert from 3-bit 2's complement number to 1's complement number

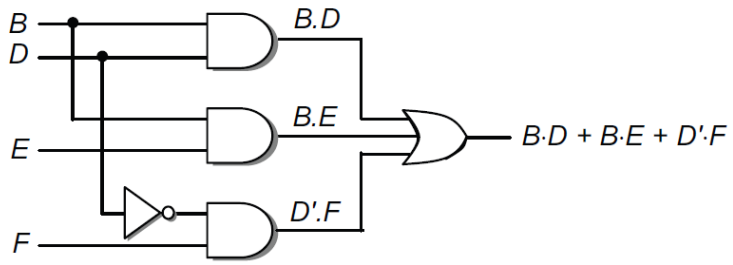
4-6. Prove that the set {AND, NOT} is a complete set of logic gates.

4-7. Draw a logic circuit for $B{\cdot}D+B{\cdot}E+D'{\cdot}F$ using only NAND gates (for the inverters as well).

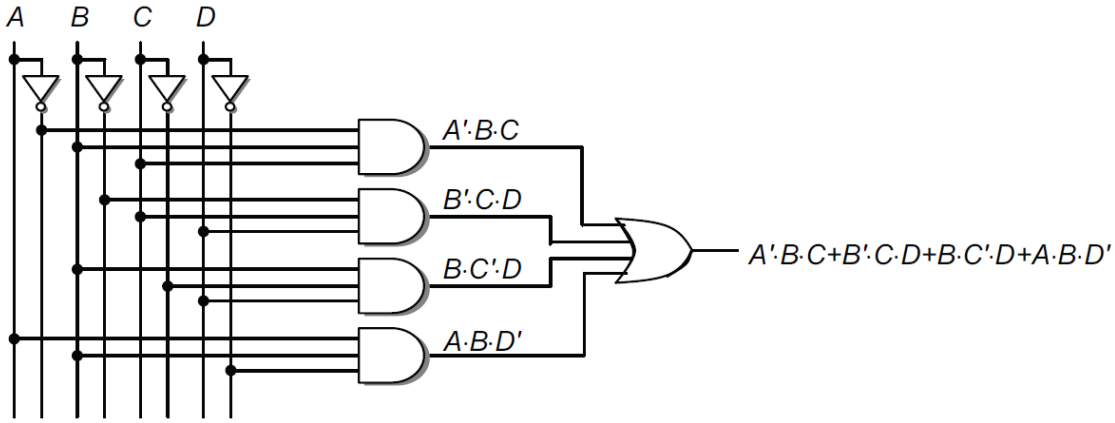
4-8. Draw a logic circuit for $(A+B+C'){\cdot}(B'+C'+D)$ using only NOR gates (for the inverters as well).

JAWABAN PART I

4-1.



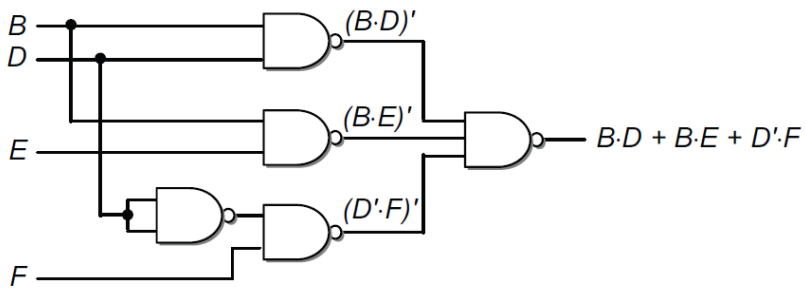
4-2.



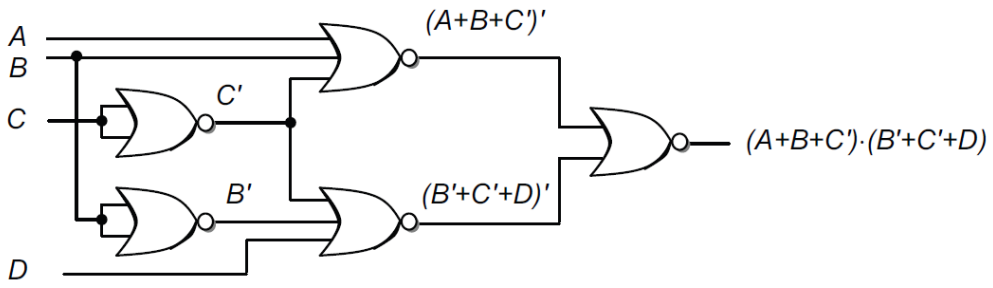
4-3. $F = x{\cdot}y+y{\cdot}z$ 4-4. (a)

4-6. It suffices to show that the OR operation can be implemented using the AND gat and inverter: $x+y = (x'{\cdot}y')'$.

4-7.

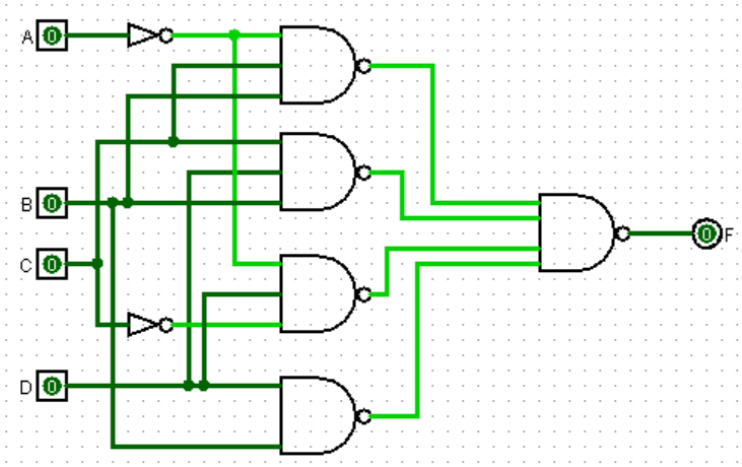


4-8.



JAWABAN PART II

1. a) Terjemahkan diagram dibawah ini kedalam bentuk persamaan boolean
   b) carilah persamaan SOPnya
   c) carilah persamaan SOP optimumnya
   d) carilah persamaan POS optimumnya.



a) F(A,B,C,D)=((A'BC)'.(BCD)'.(A'C'D)'.(BD)')'

b) F=(A'BC)+(BCD)+(A'C'D)+(BD)

c) F=(A'BC)+(BCD)+(A'C'D)+(BD)

SOP optimum: F=(BD)+(A'C'D)+(A'BC)

AB \ CD			C	
	0	1	0	0
	0	1	1	1
A	0	1	1	0
	0	0	0	0

B

D

d) F=(A'BC)+(BCD)+(A'C'D)+(BD)

Buat KMAP, didapatkan F=∑m(1,5,6,7,12,15)

Komplemen SOM = POM

F'=∑m(0,2,3,4,8,9,10,11,13,14) ≡ F Π *M*(0,2,3,4,8,9,10,11,13,14)

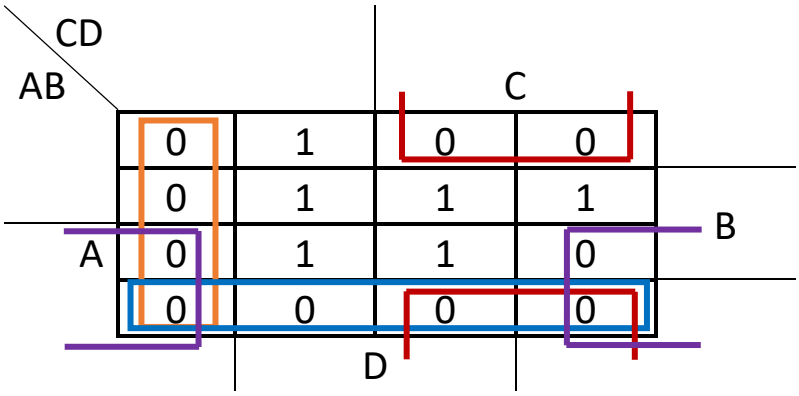
Kelompokan minterm pada F' yang berisikan nilai 0.

F'=(C'D')+(AB')+(AD')+(B'C)

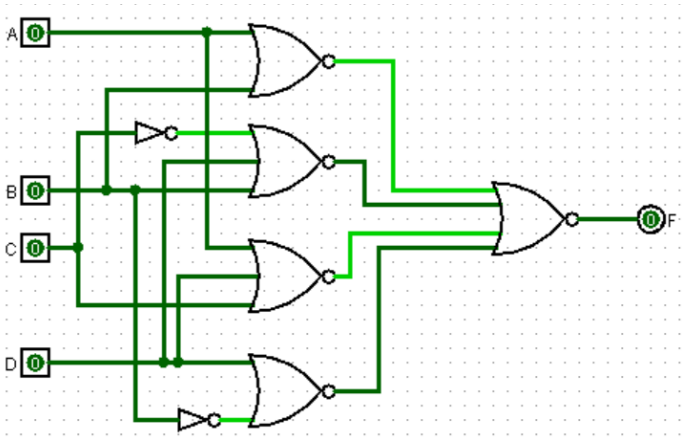
Konversi SOP ke format POS dengan Teorema DeMorgan

F=(C+D).(A'+B).(A'+D).(B+C')

Untuk verifikasi, pastikan output truth table F=(C+D).(A'+B).(A'+D).(B+C') sama dengan F(A,B,C,D)=((A'BC)'.(BCD)'.(A'C'D)'.(BD)')'



2. a)Terjemahkan diagram dibawah ini kedalam bentuk persamaan boolean
   b) carilah persamaan POS nya
   c) carilah persamaan POS optimumnya
   d) carilah persamaan SOP optimumnya.



a) F(A,B,C,D)=((A+B)' + (B+C'+D)' + (A+C+D)' + (B'+D)')'

b) F=(A+B).(B+C'+D).(A+C+D).(B'+D)

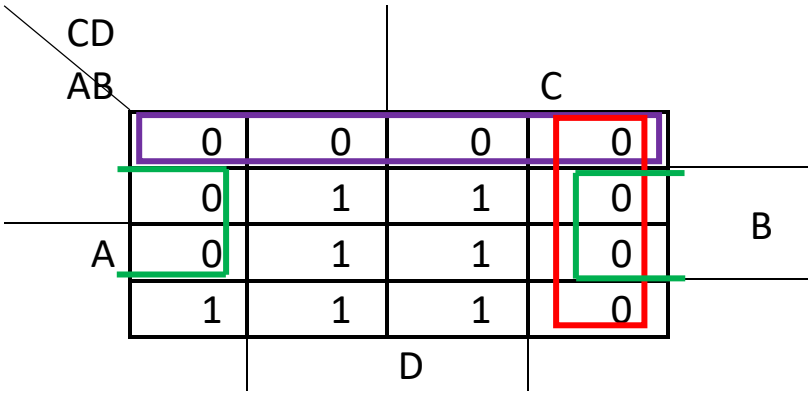
c) F=(A+B).(B+C'+D).(A+C+D).(B'+D)

Konversi POS ke format SOP dengan mengimplementasikan teorema DeMorgan

F'=((A+B).(B+C'+D).(A+C+D).(B'+D))'

F'=A'B'+B'CD'+A'C'D'+BD'

Terjemahkan persamaan pada KMAP dengan mengelompokkan kotak berisikan 0.



F'=A'B'+CD'+BD'

Konversi SOP ke format POS dengan Teorema DeMorgan

F=(A+B).(C'+D).(B'+D)

d) F=(A+B).(B+C'+D).(A+C+D).(B'+D)

Konversi POS ke format SOP dengan mengimplementasikan teorema DeMorgan
F'=((A+B).(B+C'+D).(A+C+D).(B'+D))'
F'=A'B'+B'CD'+A'C'D'+BD'

Terjemahkan persamaan dengan mengisi kotak KMAP dengan 0, mengingat POM merupakan komplemen SOM. Dari KMAP didapatkan
F'=$\sum$m(0,1,2,3,4,6,10,12,14) ≡ F Π *M*(0,1,2,3,4,6,10,12,14)

Mengingat SOM merupakan komplemen POM sehingga komplemen dari F'=$\sum$m(0,1,2,3,4,6,10,12,14) → F=$\sum$m(5,7,8,9,11,13,15) (ditandai dengan kotak bernilai 1).

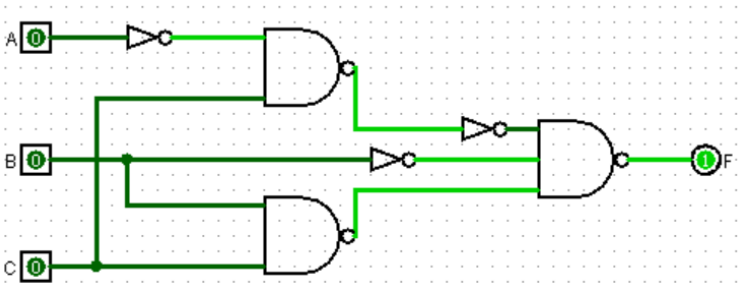
Selanjutnya kelompokkan minterms, hasil SOP optimum: F(A,B,C,D) =BD+AD+AB'C'

Untuk verifikasi, pastikan output truth table F(A,B,C,D) =BD+AD+AB'C' sama dengan F(A,B,C,D)=((A+B)' + (B+C'+D)' + (A+C+D)' + (B'+D)')'

AB \ CD				
	0	0	0	0
	0	1	1	0
A	0	1	1	0
	1	1	1	0

C, B, D

3. a) Apakah persamaan boolean dari circuit dibawah ini
   b) Apakah output fungsi F jika input A=1, B=0 dan C=1 ?
   c) Ubahlah circuit hanya dengan gate OR dan Invert

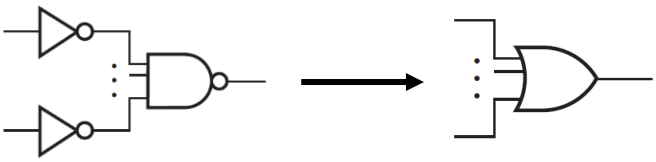


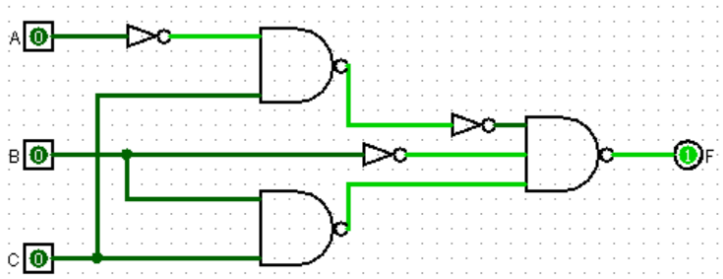
a) F=((A'C).B'.(BC)')'

b) F= 1

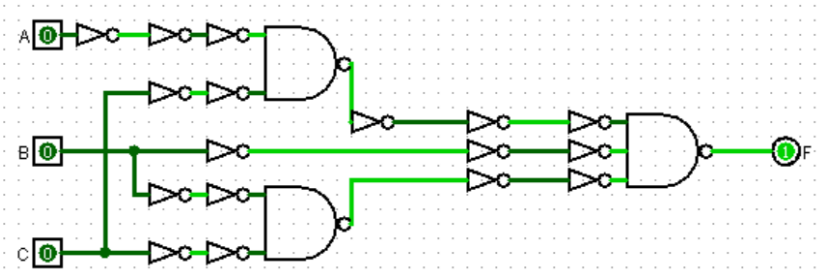
A	B	C	F
0	0	0	**1**
0	0	1	**0**
0	1	0	**1**
0	1	1	**1**
1	0	0	**1**
1	0	1	**1**
1	1	0	**1**
1	1	1	**1**

c) NAND ≡ NEG-OR
NEG-NAND ≡ NEG-NEG-OR
NEG-NAND ≡ OR

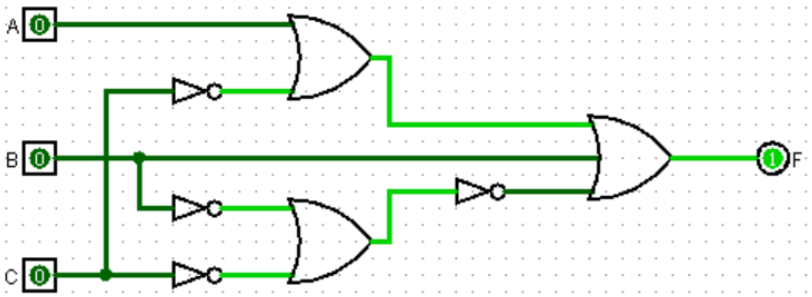




Tambahkan dua negasi sebelum NAND

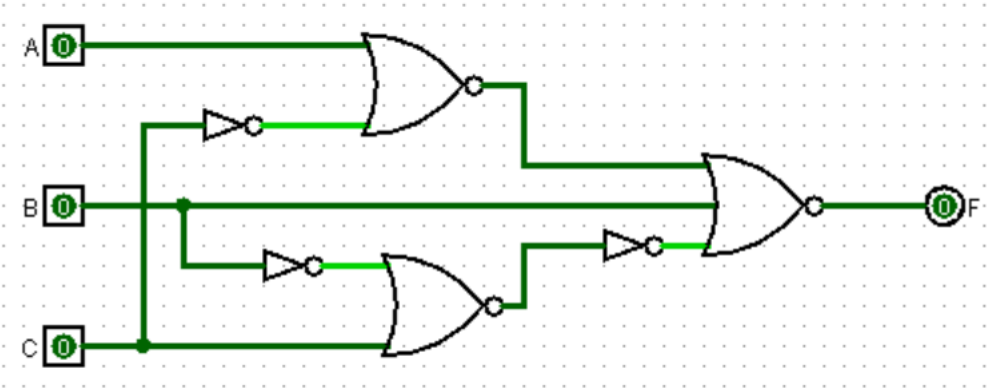


Mengingat NEG-NAND ≡ OR, ganti gate NEG-NAND dengan gate OR.
Hilangkan dua inverter yang berada dalam satu jalur karena ((X)')' = X



F=((A'C).B'.(BC)')' = ((A'C).B'.(B'+C'))'=(A+C')+B+(B'+C')'

4. a) Apakah persamaan boolean dari circuit dibawah ini
   b) Apakah output fungsi F jika input A=0, B=1 dan C=1 ?
   c) Ubahlah circuit hanya dengan gate AND dan Invert

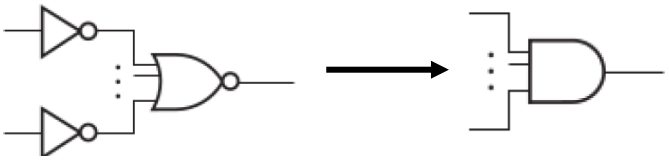


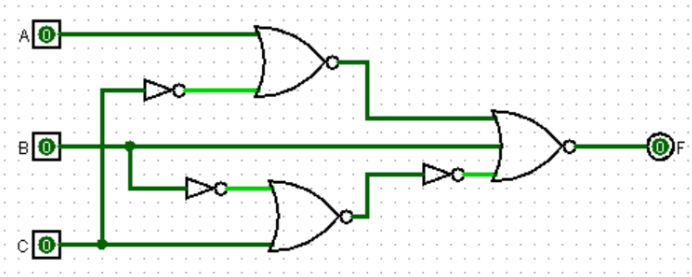
a) F=((A+C')'+B+(B'+C))'

b) F= 0

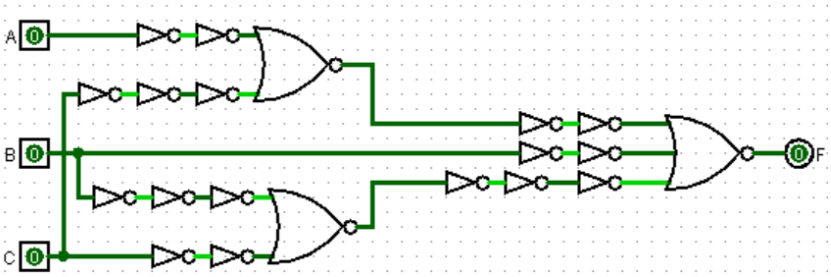
A	B	C	X
0	0	0	**0**
0	0	1	**0**
0	1	0	**0**
0	1	1	**0**
1	0	0	**0**
1	0	1	**0**
1	1	0	**0**
1	1	1	**0**

c) NOR ≡ NEG-AND
NEG-NOR ≡ NEG-NEG-AND
NEG-NOR ≡ AND

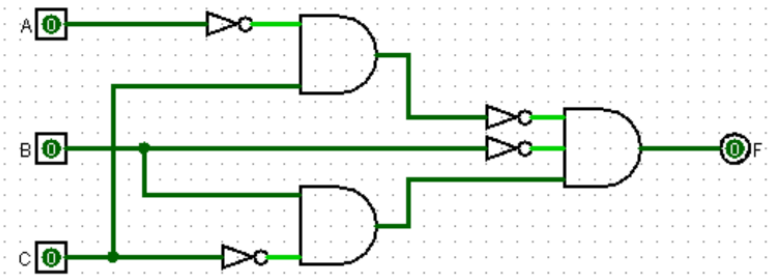




Tambahkan dua negasi sebelum NOR



Mengingat NEG-NOR ≡ AND, ganti gate NEG-NOR dengan gate AND.
Hilangkan dua inverter yang berada dalam satu jalur karena ((X)')' = X



F=((A+C')'+B+(B'+C))' = (A'C)'.B'.(BC')

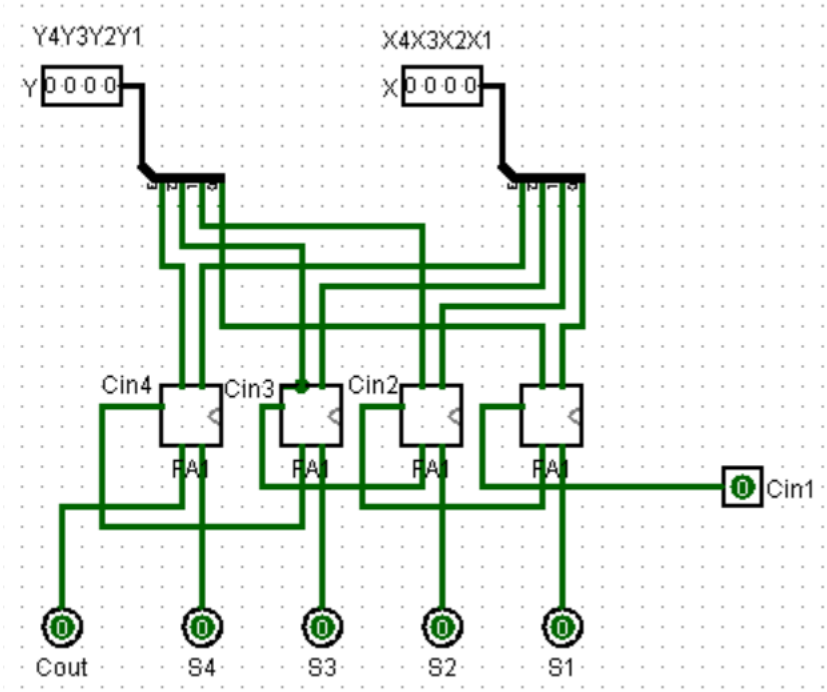
5. Ubah F= C'D + AB'C + BD' ke persamaan yang hanya menggunakan NAND dan Invert

   F=((C'D)'.(AB'C)'.(BD')')'

6. Ubah F= (A'+B'+C).(B+C').D ke persamaan yang hanya menggunakan NOR dan Invert

   F=((A'+B'+C)'+(B+C')'+D')'

7. Jika input Y4Y3Y2Y1=0011 ; X4X3X2X1=1011; Cin1=1, maka berapakah nilai S4S3S2S1 dan Cout?



S4S3S2S1=1111; Cout=0